

PC lab 8

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Exercise 01

```
manuth@manuth-LOQ-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise01$ OMPI_MCA_btl=self,sm mpirun -np 2 ./job1
Hello World! From rank 0 machine manuth-LOQ-15IRH8
Just a normal process From rank 1 machine manuth-LOQ-15IRH8
manuth@manuth-LOQ-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise01$ OMPI_MCA_btl=self,sm mpirun -np 4 ./job1
Just a normal process From rank 1 machine manuth-LOQ-15IRH8
Hello World! From rank 0 machine manuth-LOQ-15IRH8
Just a normal process From rank 3 machine manuth-LOQ-15IRH8
Just a normal process From rank 2 machine manuth-LOQ-15IRH8
manuth@manuth-LOQ-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise01$
```

The program was executed using four MPI processes. Each process received a unique rank from 0 to 3. Rank 0 printed the “Hello World” message, while the remaining ranks printed the normal process message.

Exercise 02

```
manuth@manuth-LOQ-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab$ cd Exercise02
manuth@manuth-LOQ-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise02$ nano sum_mpi.cpp
manuth@manuth-LOQ-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise02$ mpic++ -o sum_mpi sum_mpi.cpp
manuth@manuth-LOQ-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise02$ OMPI_MCA_btl=self,sm mpirun -np 4 ./sum_mpi
Rank 3: 7500001 to 10000000 -> Local sum = 21875001250000
Rank 1: 2500001 to 5000000 -> Local sum = 9375001250000
Rank 2: 5000001 to 7500000 -> Local sum = 15625001250000
Rank 0: 1 to 2500000 -> Local sum = 3125001250000
Total sum = 500000005000000
manuth@manuth-LOQ-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise02$
```

```

manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise
02$ nano sum_mpi.cpp
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise
02$ mpic++ -o sum_mpi sum_mpi.cpp
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise
02$ OMPI_MCA_btl=self,sm mpirun -np 4 ./sum_mpi
Rank 2: 5000001 to 7500000 -> Local sum = 15625001250000
Rank 3: 7500001 to 10000000 -> Local sum = 21875001250000
Rank 1: 2500001 to 5000000 -> Local sum = 9375001250000
Rank 0: 1 to 2500000 -> Local sum = 3125001250000
Total sum = 50000005000000
Execution time = 0.00403013 seconds
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise
02$ █

```

Exercise 03

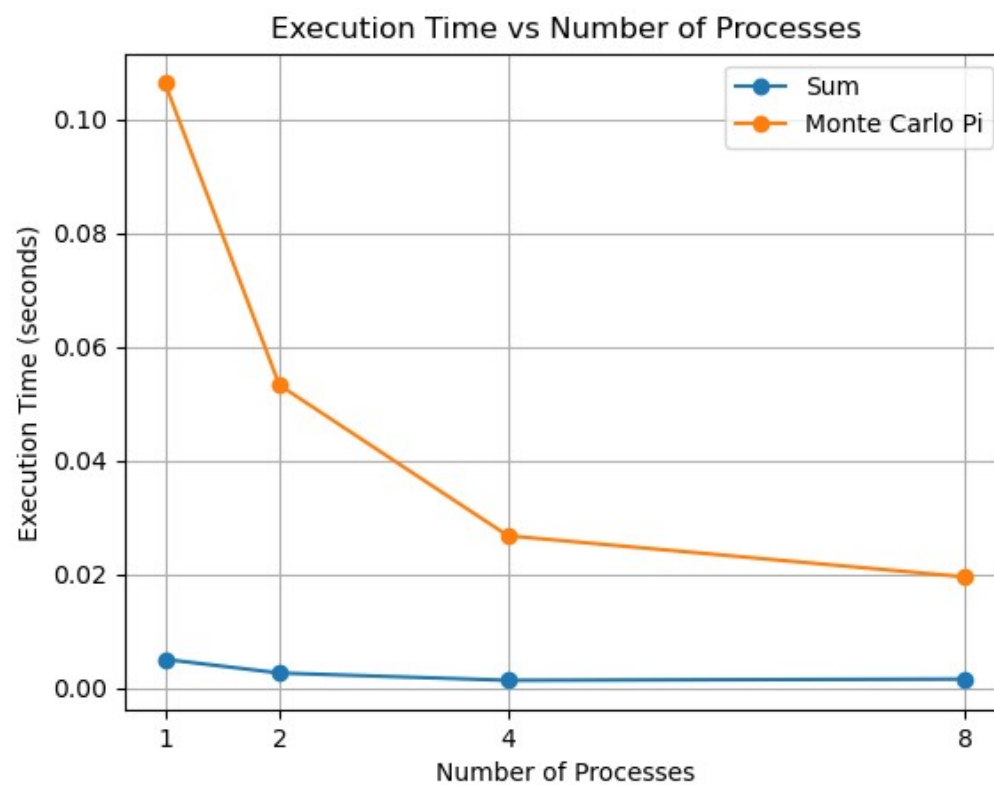
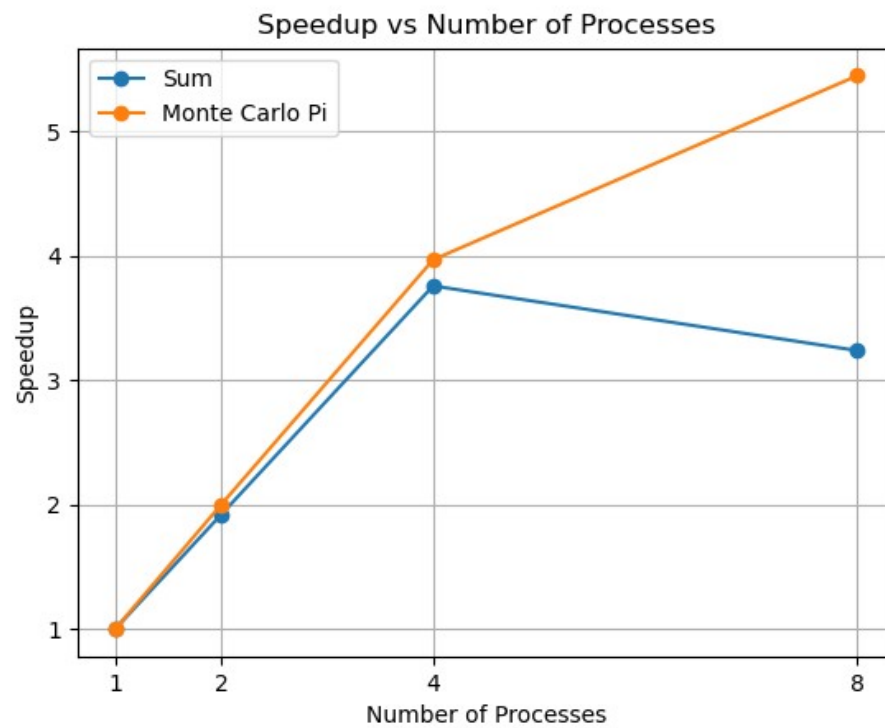
MPI Monte Carlo calculation of π using 4 processes

```

manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise
03$ nano pi_mpi.cpp
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise
03$ mpic++ -o pi_mpi pi_mpi.cpp
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise
03$ OMPI_MCA_btl=self,sm mpirun -np 4 ./pi_mpi
Points inside circle = 7851912
Calculated Pi = 3.14076480
Execution time = 0.04122250 seconds
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise
03$ █

```

Exercise 04



Exercise 05

```
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/Exercise05$  
nano mismatch.cc  
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/Exercise05$  
mpic++ -o mismatch mismatch.cc  
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/Exercise05$  
timeout 5s env OMPI_MCA_btl=self,sm mpirun -np 3 ./mismatch  
Rank 0 sending value to Rank 1  
Rank 1 waiting for message from Rank 2...  
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/Exercise05$
```

```
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise  
05$ nano bsend.cc  
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise  
05$ mpic++ -o bsend bsend.cc  
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise  
05$ OMPI_MCA_btl=self,sm mpirun -np 2 ./bsend  
Rank 0 buffered sending value 100 to Rank 1  
Rank 1 received value 100  
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise  
05$
```

Exercise 06

```
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab$ mkdir E  
xercise06  
cd Exercise06  
nano any_source.cc  
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise  
06$ mpic++ -o any_source any_source.cc  
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise  
06$ OMPI_MCA_btl=self,sm mpirun -np 3 ./any_source  
Rank 0 sent 100 to Rank 2  
Rank 1 sent 200 to Rank 2  
Rank 2 received 100 from Rank 0
```

Exercise 07

```
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise  
07$ mpic++ -o any_source_bsend any_source_bsend.cc  
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise  
07$ OMPI_MCA_btl=self,sm mpirun -np 3 ./any_source_bsend  
Rank 0 buffered sent 100 to Rank 2  
Rank 1 buffered sent 200 to Rank 2  
Rank 2 received 100 from Rank 0  
Rank 2 received 200 from Rank 1  
manuth@manuth-L0Q-15IRH8:/run/media/manuth/New Volume2/sliit/Y3S1/new/PC/LAB/lab8/lab/Exercise  
07$ █
```